Height Growth Equations for CONUS: Red Alder

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Data

We took the FIA height growth data subset for red alder, filtered out FIA plot locations with missing EPA L3 Codes, and where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 7702 growth observations.

The observations are distributed among L3 Codes as follows:

NA_L3NAME	NA_L3CODE	n
Coast Range	7.1.8	4253
Cascades	6.2.7	1724
North Cascades	6.2.5	617
Strait of Georgia/Puget Lowland	7.1.7	588
NA	NA	212
Klamath Mountains	6.2.11	158
Willamette Valley	7.1.9	89
Columbia Mountains/Northern Rockies	6.2.3	28
Eastern Cascades Slopes and Foothills	6.2.8	14
Blue Mountains	6.2.9	13
Sierra Nevada	6.2.12	5
California Coastal Sage, Chaparral, and Oak Woodlands	11.1.1	1

Height Growth Prediction

We fit the equation:

$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl + \beta_5 PPT_{at} + \beta_6 PPT_{sm} + \beta_7 EMT)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \tag{1}$$

to the all 7702 growth observations, estimating β_0 through β_6. Where:

- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- PPT_{at} = autumn precipitation (mm)
- PPT_{sm} = summer precipitation (mm)
- EMT = extreme minimum temperature ($^\circ C$)

ht_{max} was estimated by finding the maximum height observed in the data set for red alder. In our data that value is: 151 feet.

The parameter estimates and fit statistics are:

	Coef.	Std. error	t-stat.	p
b1	0.0241733	0.0011652	20.746876	0.0000000
b2	2.0127781	0.1417486	14.199637	0.0000000
b3	-0.0441968	0.0279531	-1.581103	0.1138955
b4	0.0016205	0.0001484	10.917027	0.0000000
b5	0.0005087	0.0000696	7.310327	0.0000000
b6	-0.0023436	0.0002861	-8.191382	0.0000000
b7	0.0032893	0.0024764	1.328280	0.1841251

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Residual Standard Error: 8.16307817444514 on 7695 degrees of freedom. AIC =  54208.9
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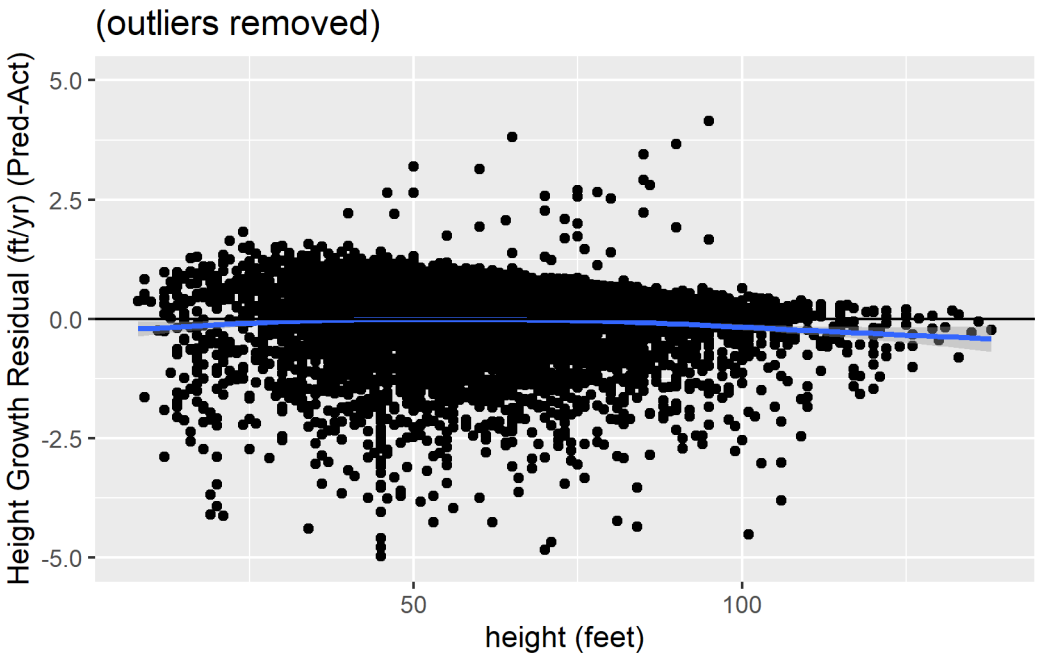
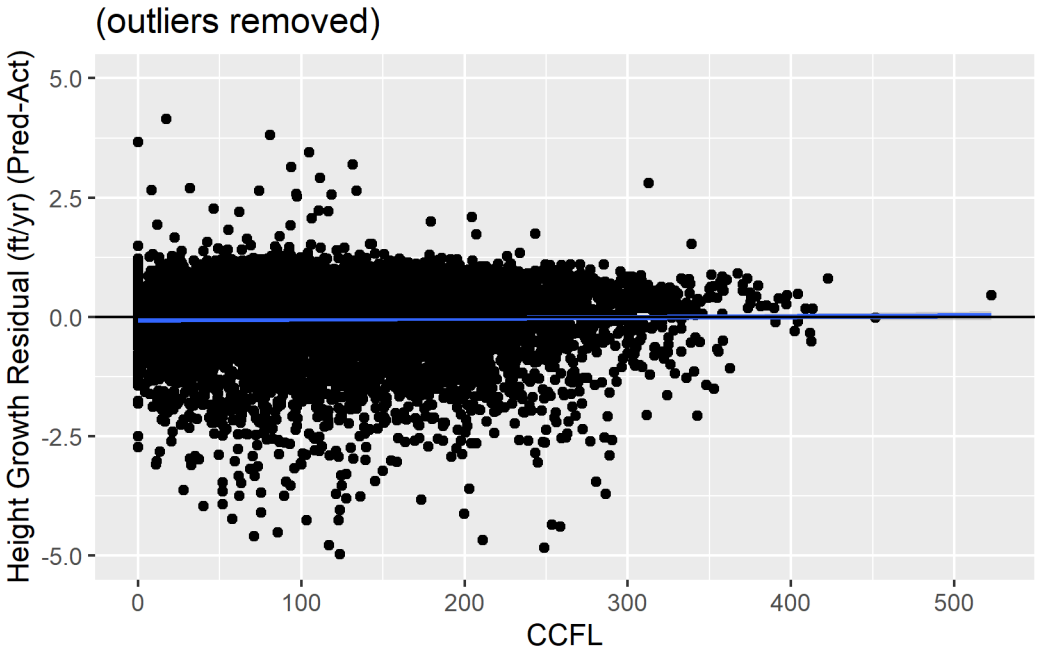
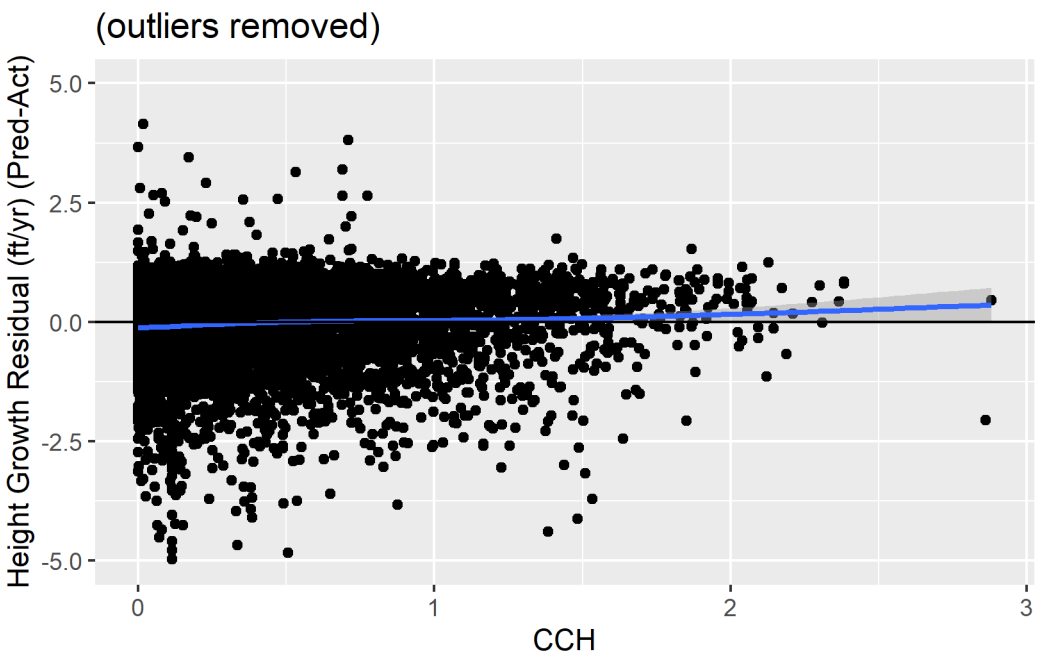
Residual Analysis for Equation 1

Mensurational Variables





(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
0.25
0.50
0.75
1.00
cr (fraction of ht)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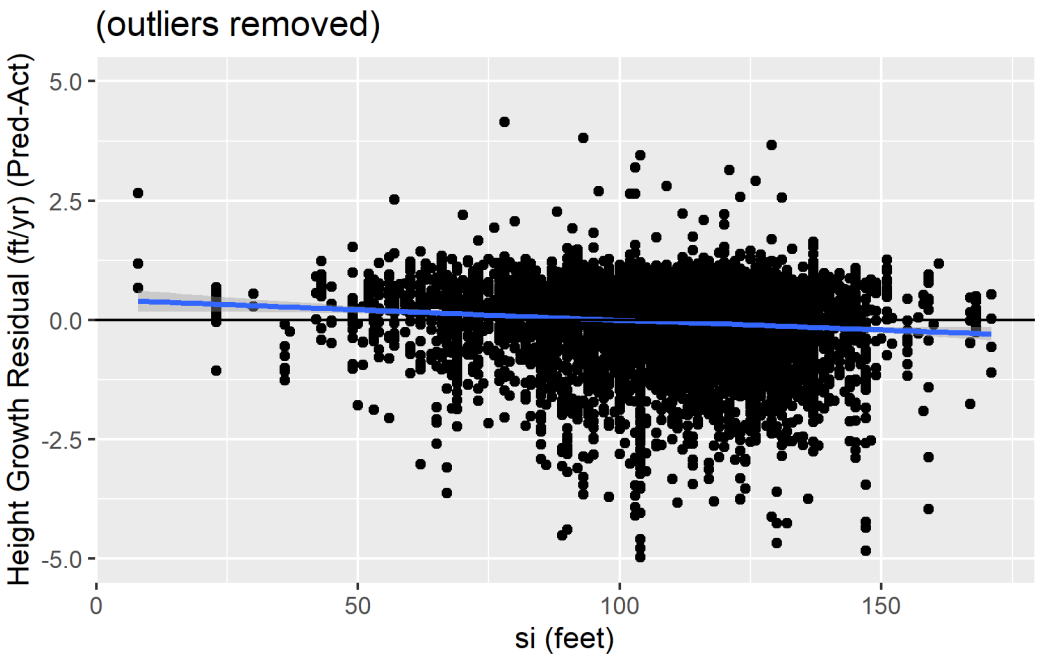
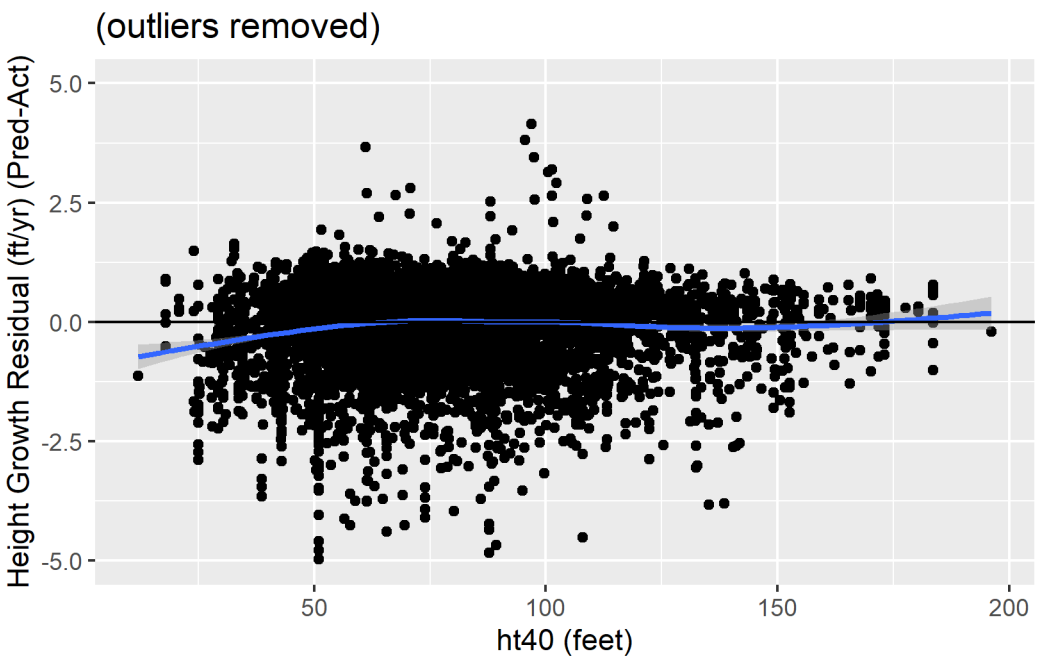
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
0
1
2
3
CCH


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
0
50
100
150
si (feet)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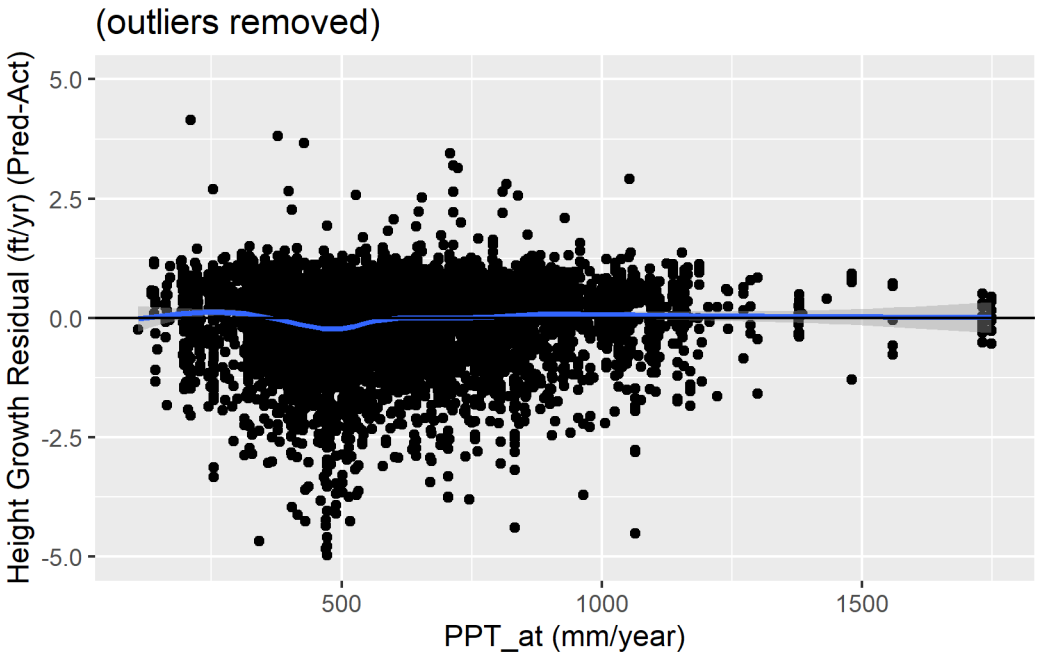
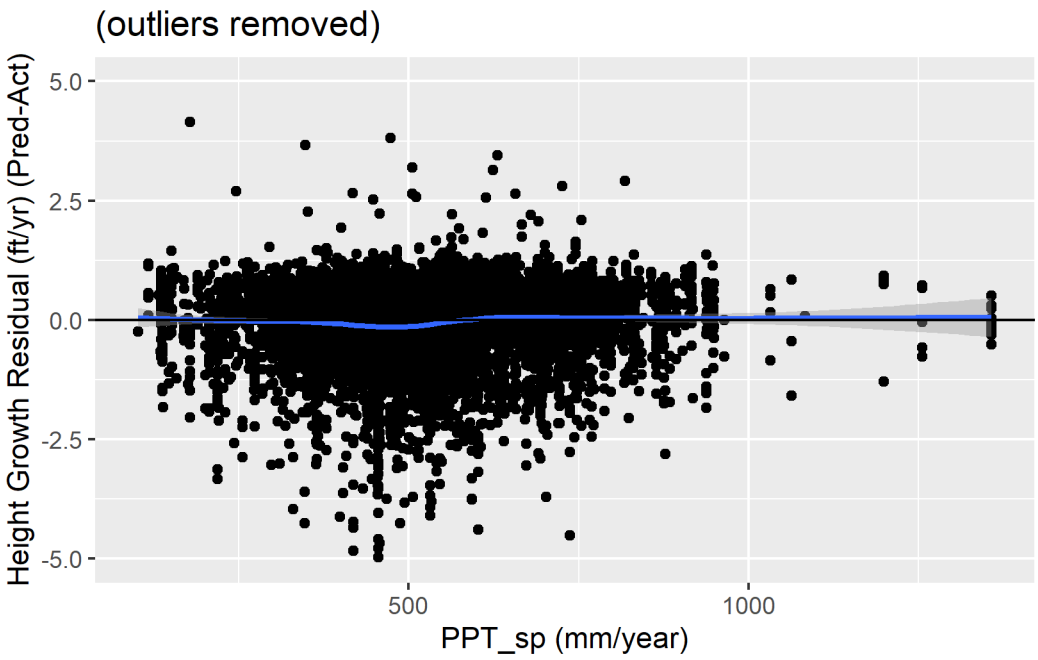
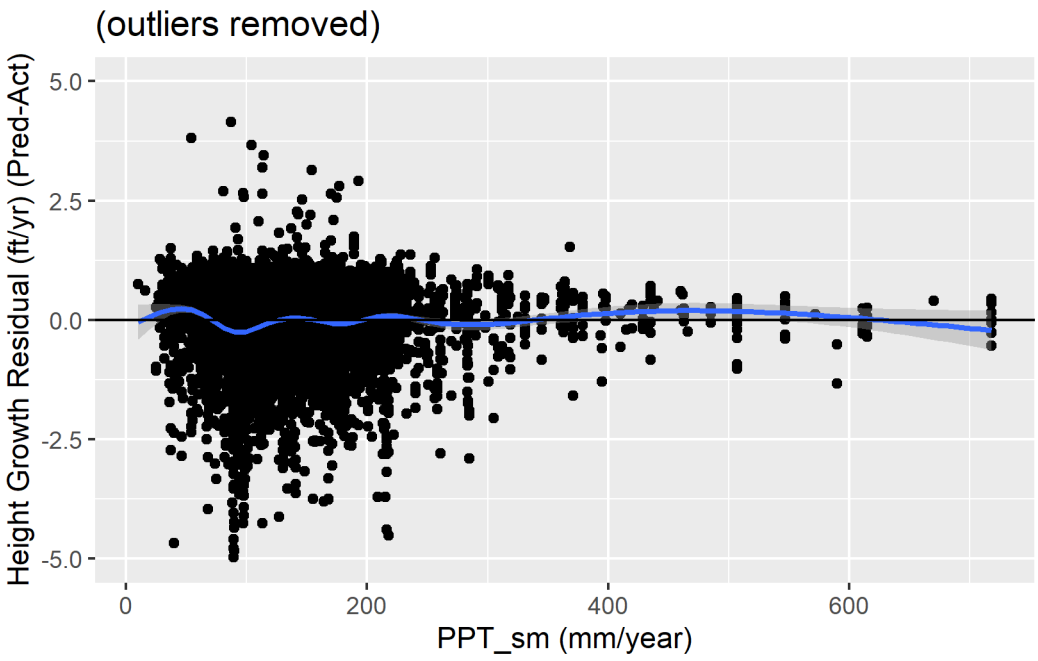
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
50
100
150
200
ht40 (feet)

Environmental and Climate Variables



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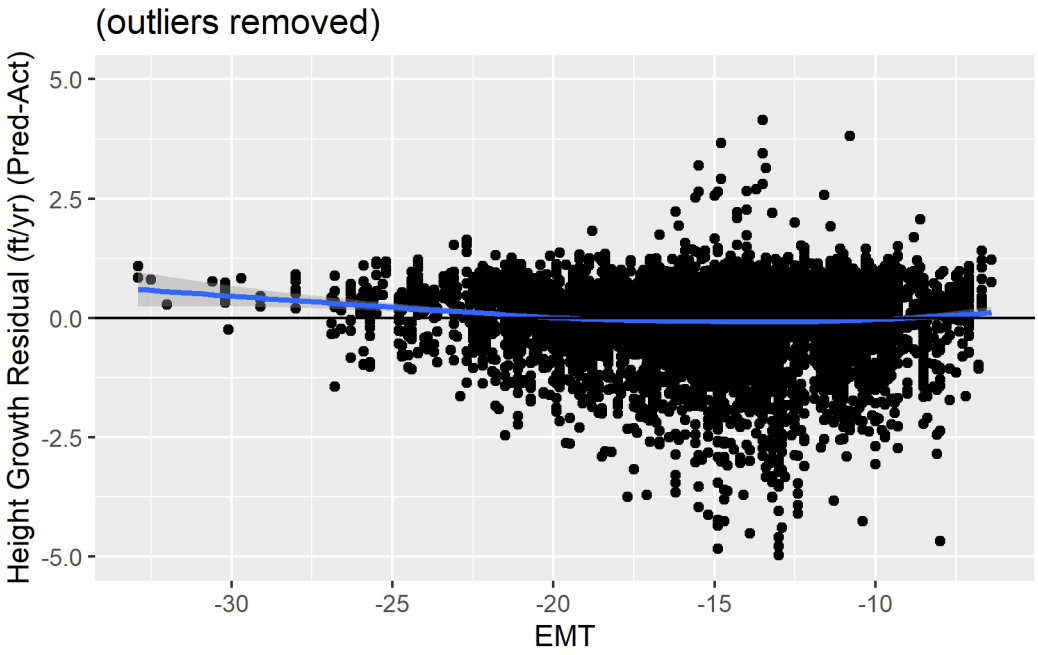
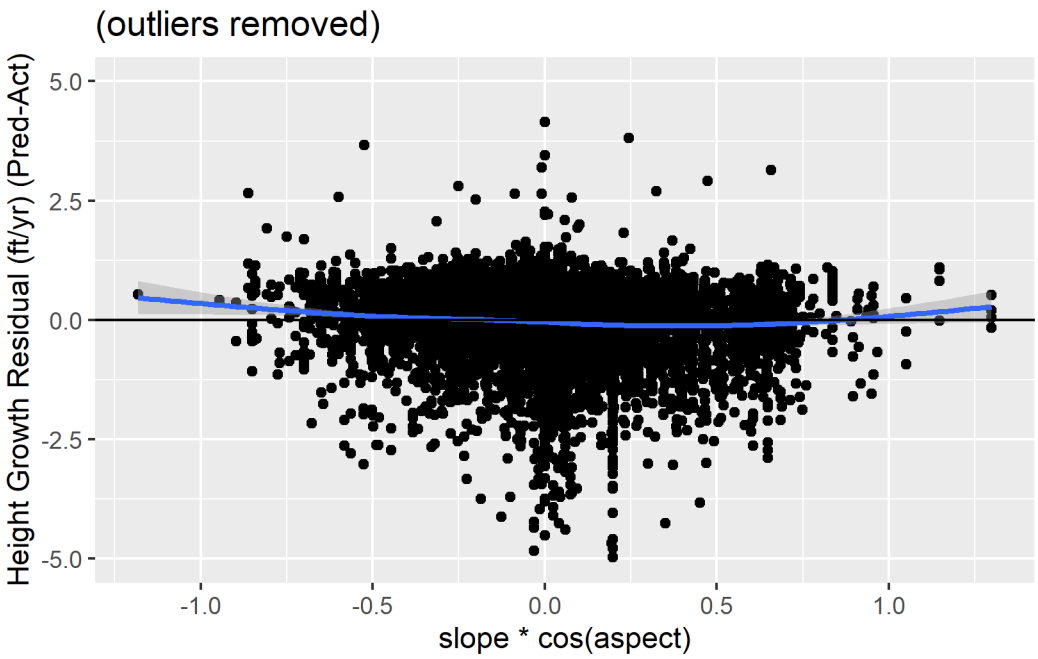
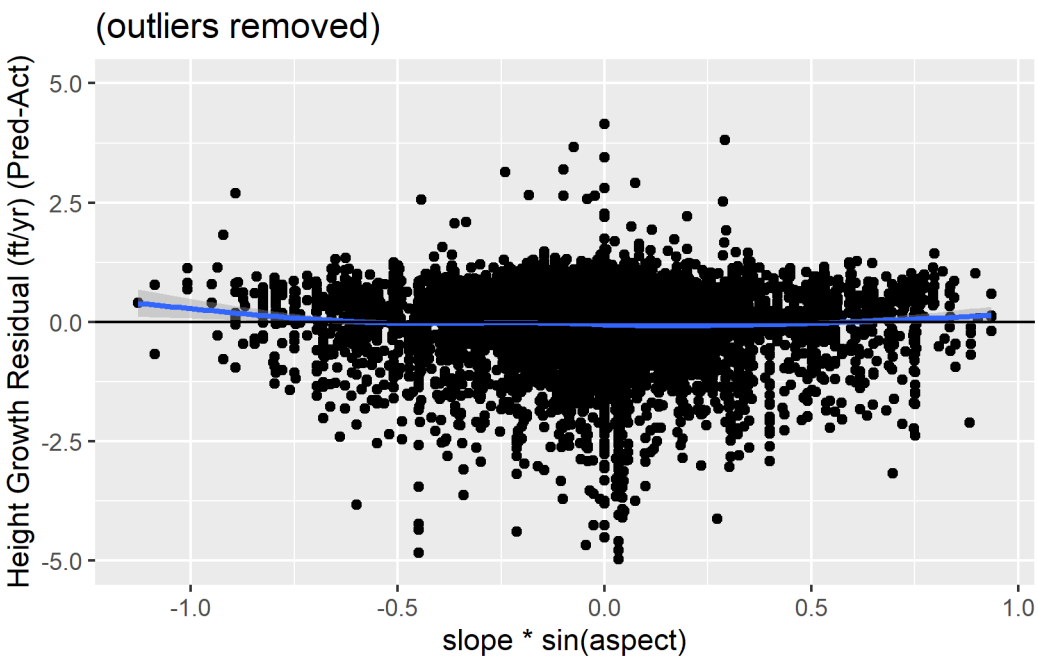
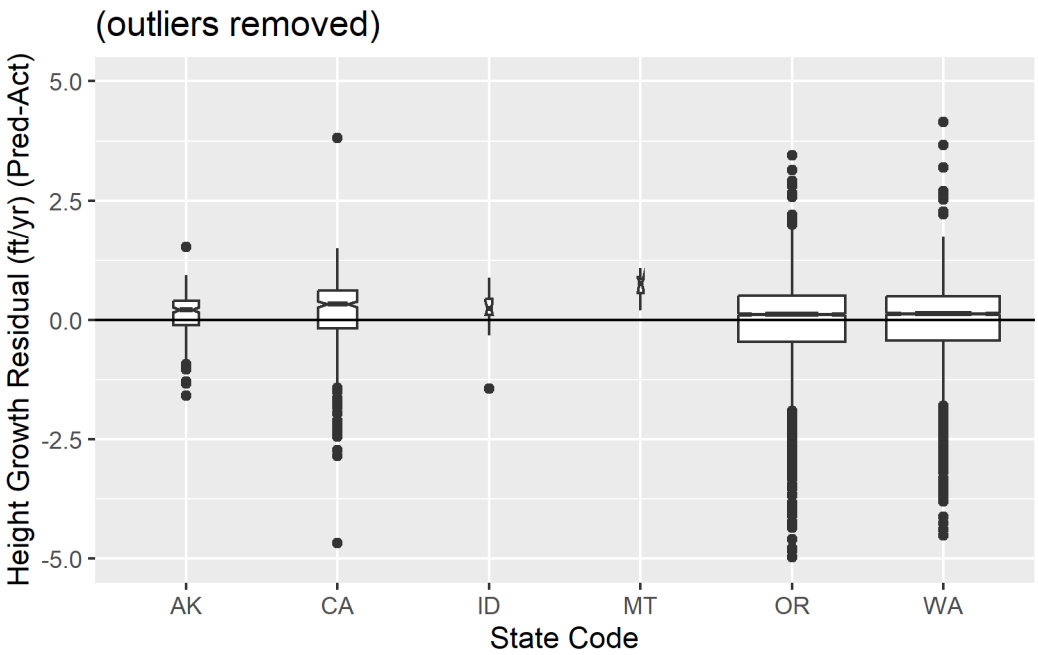
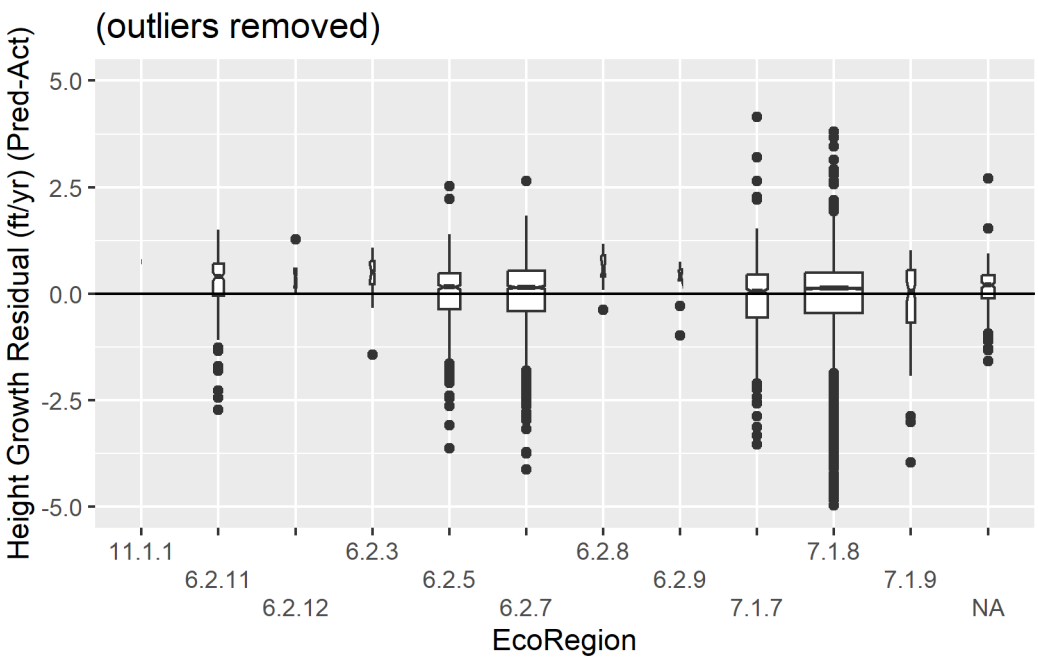
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
0
200
400
600
PPT_sm (mm/year)

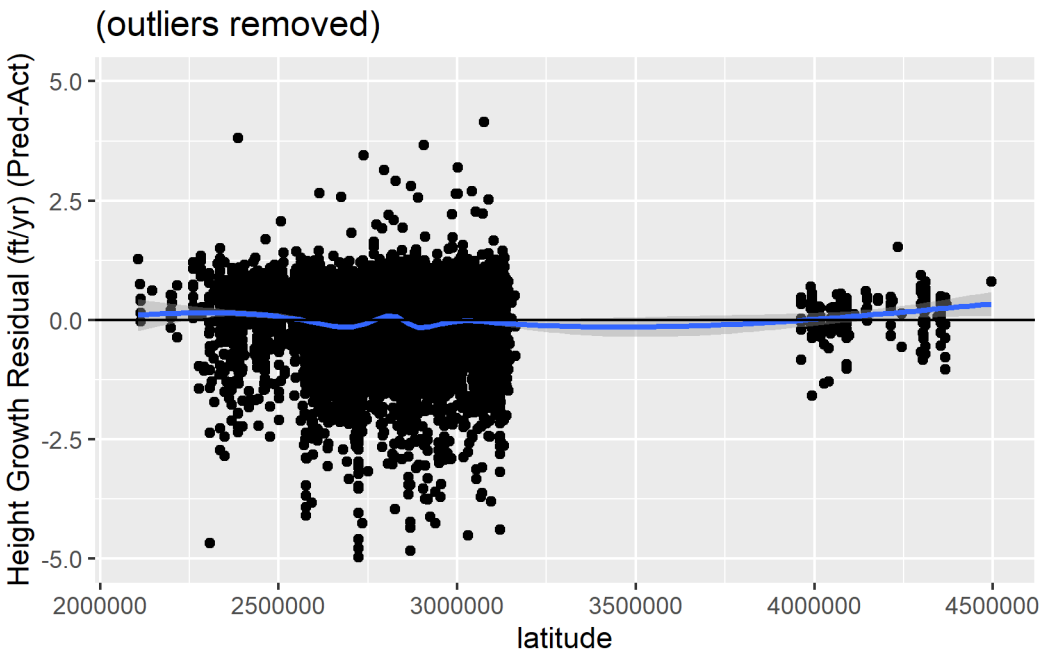

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
-30
-25
-20
-15
-10
EMT

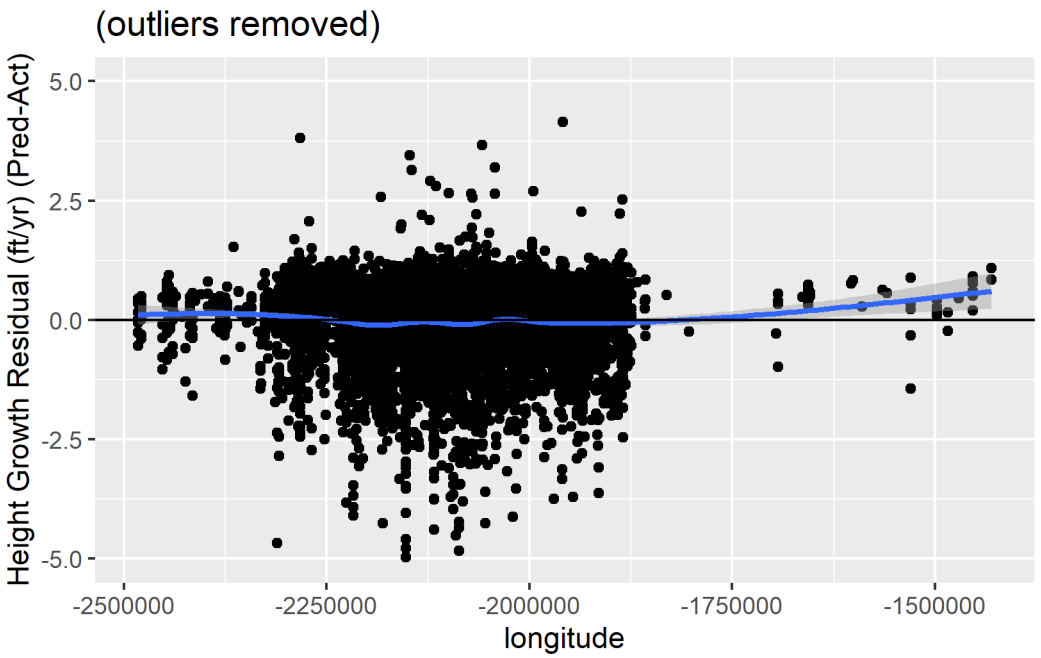

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
-1.0
-0.5
0.0
0.5
1.0
slope * cos(aspect)

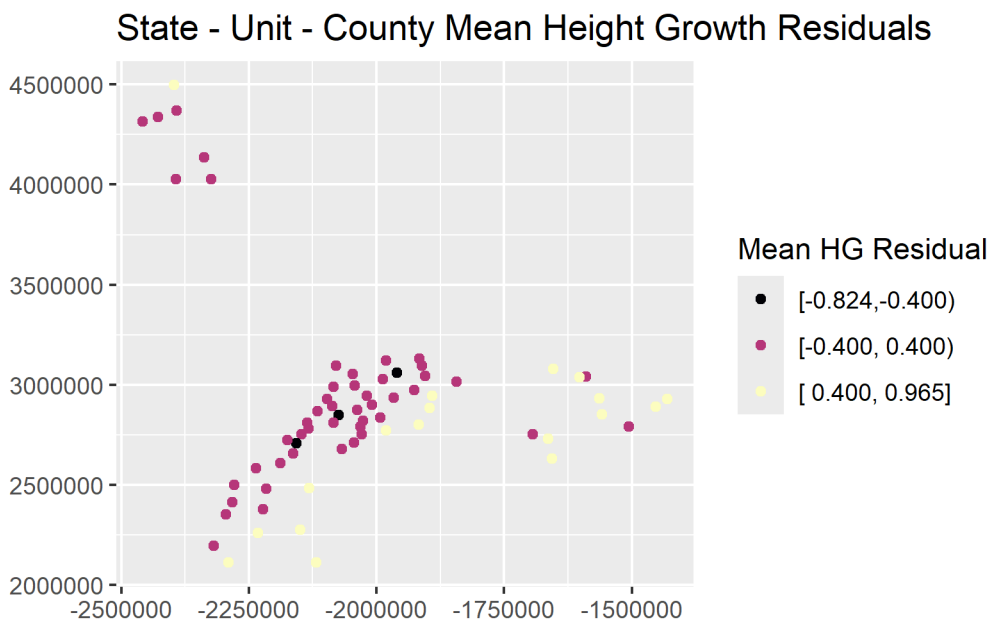

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
-1.0
-0.5
0.0
0.5
1.0
slope * sin(aspect)


(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
AK
CA
ID
MT
OR
WA
State Code

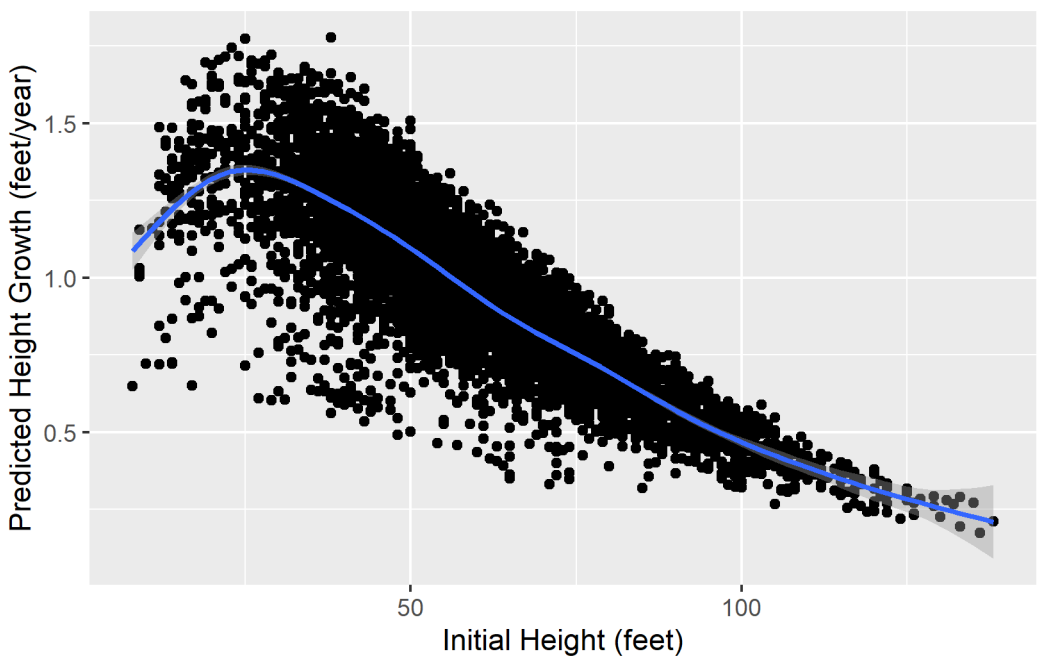

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
11.1.1
6.2.11
6.2.12
6.2.3
6.2.5
6.2.7
6.2.8
6.2.9
7.1.7
7.1.8
7.1.9
NA
EcoRegion

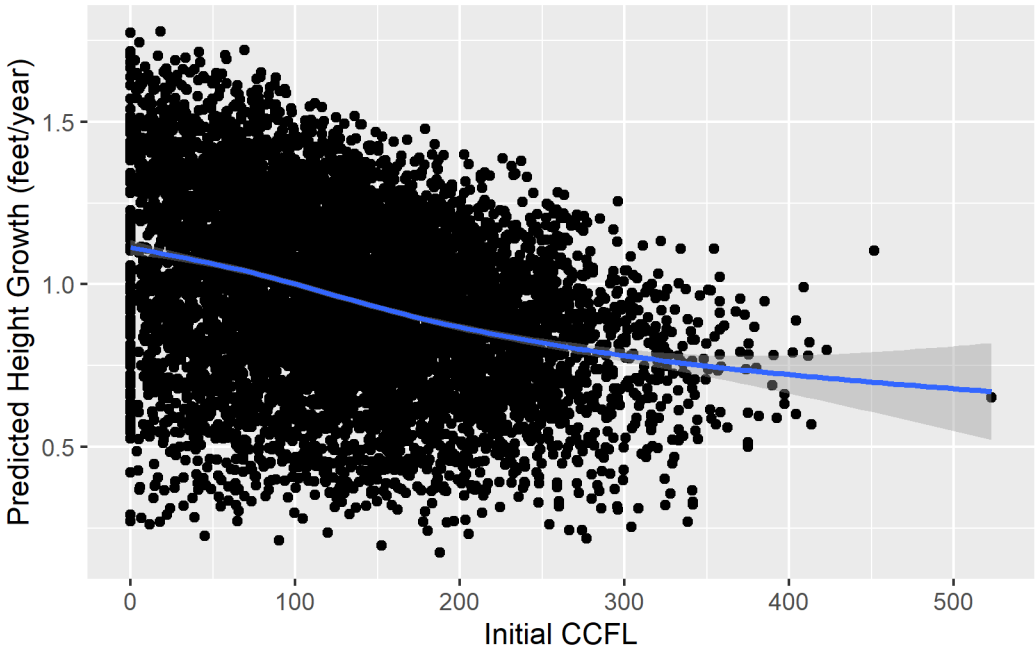

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
2000000
2500000
3000000
3500000
4000000
4500000
latitude

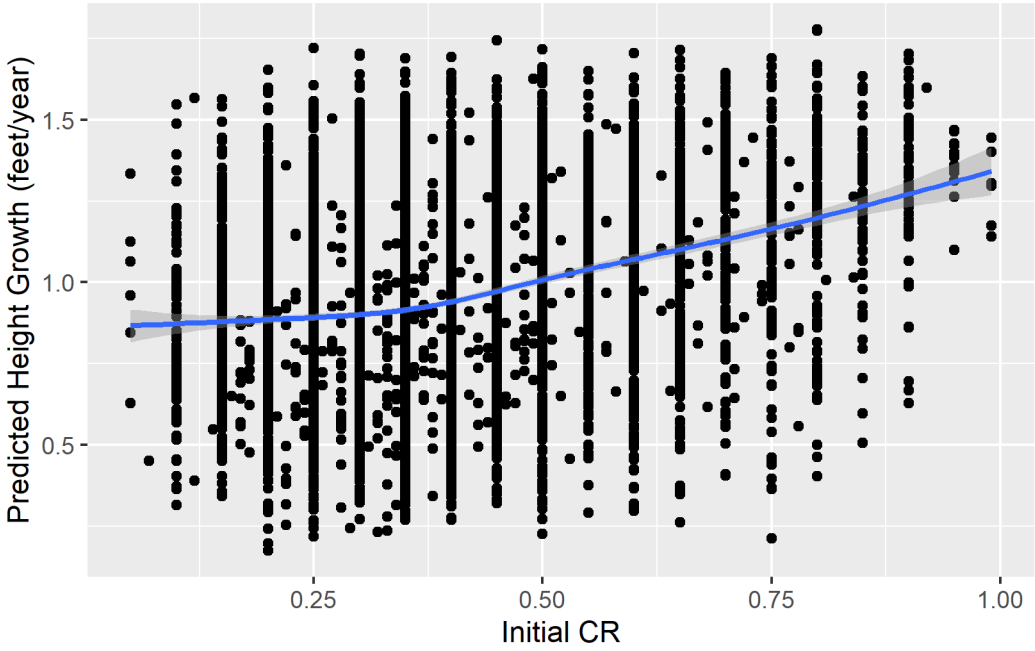

(outliers removed)
Height Growth Residual (ft/yr) (Pred-Act)
5.0
2.5
0.0
-2.5
-5.0
-2500000
-2250000
-2000000
-1750000
-1500000
longitude

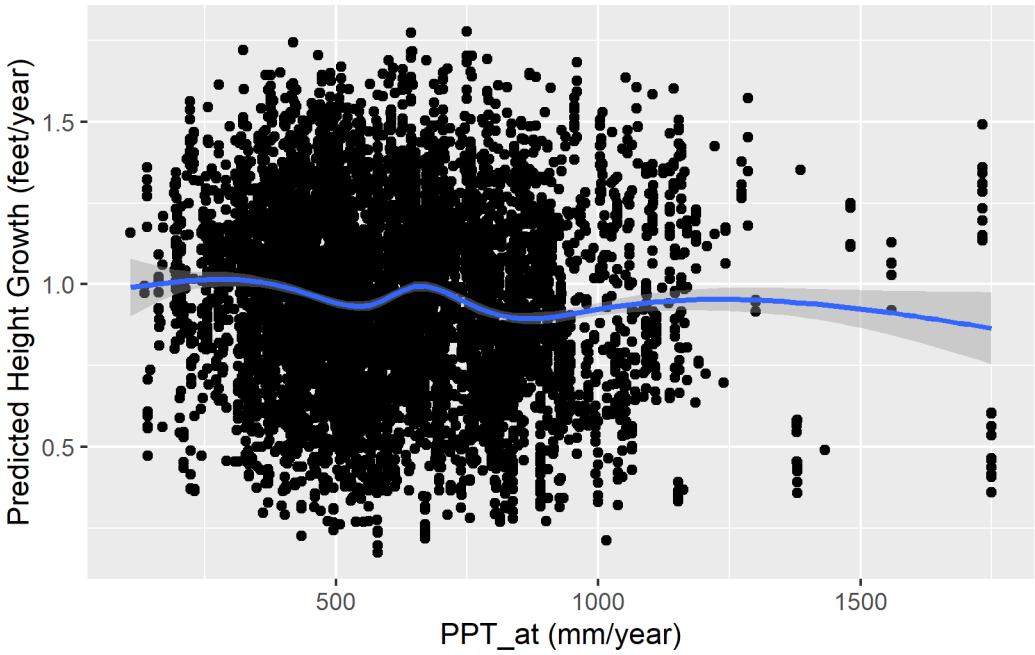


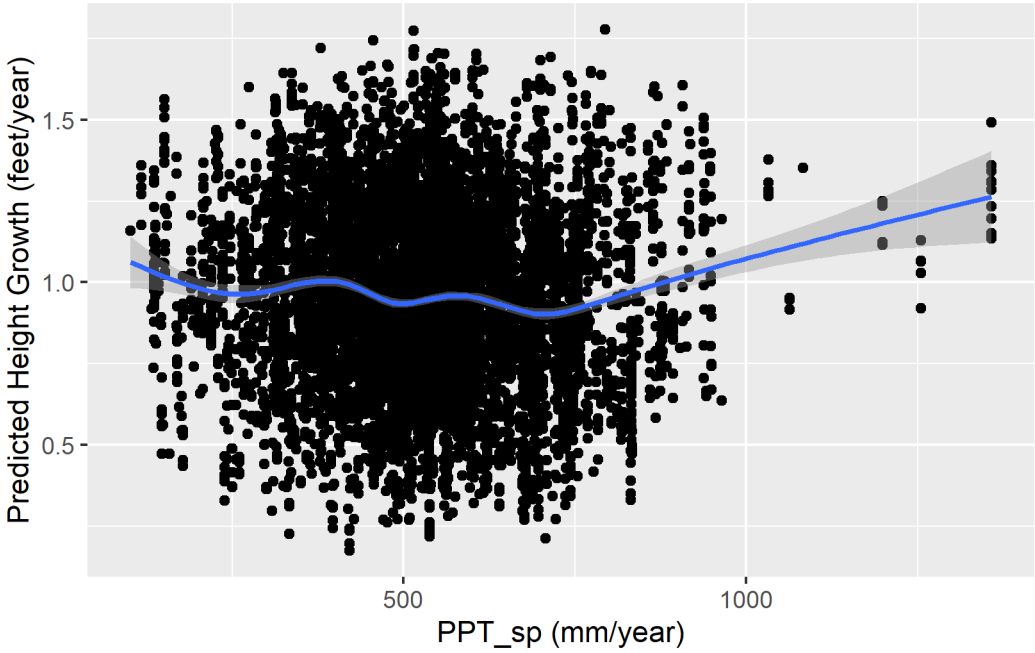
Prediction Performance

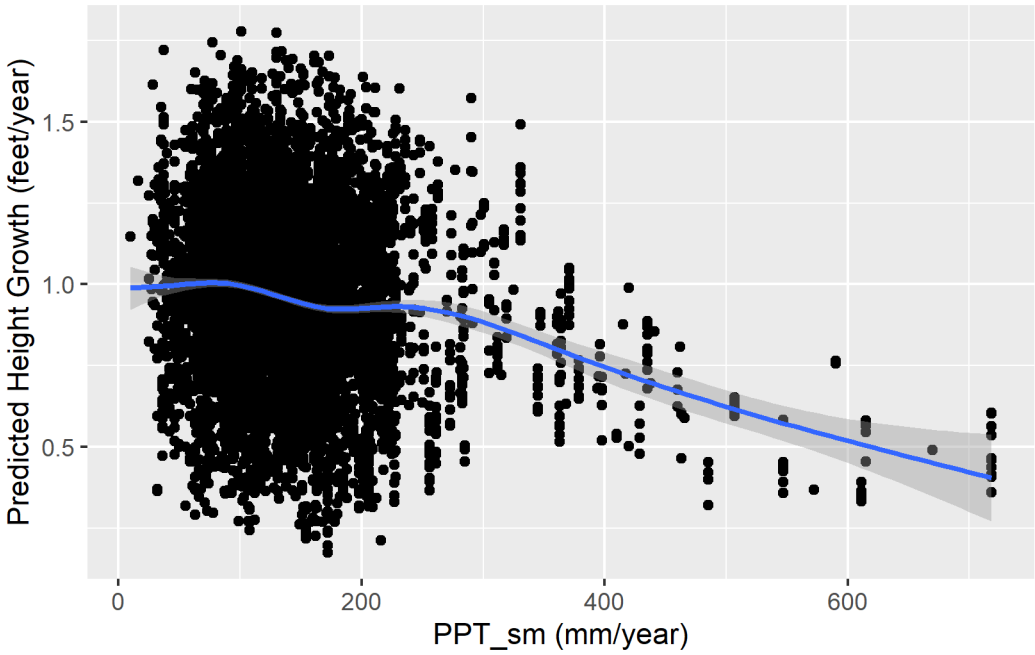


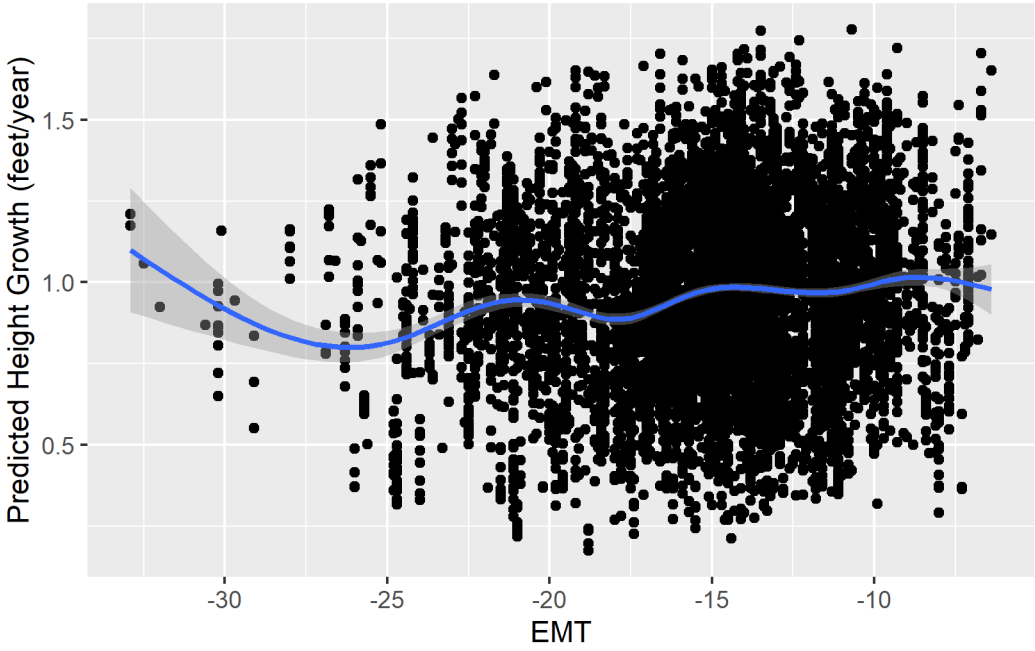

Predicted Height Growth (feet/year)
1.5
1.0
0.5
0
100
200
300
400
500
Initial CCFL


Predicted Height Growth (feet/year)
1.5
1.0
0.5
0.25
0.50
0.75
1.00
Initial CR


Predicted Height Growth (feet/year)
1.5
1.0
0.5
500
1000
1500
PPT_at (mm/year)


Predicted Height Growth (feet/year)
1.5
1.0
0.5
500
1000
PPT_sp (mm/year)


Predicted Height Growth (feet/year)
1.5
1.0
0.5
0
200
400
600
PPT_sm (mm/year)


Predicted Height Growth (feet/year)
1.5
1.0
0.5
-30
-25
-20
-15
-10
EMT